

Building a Case

Module 2 · Stationarity as a Verdict, Not a Measurement

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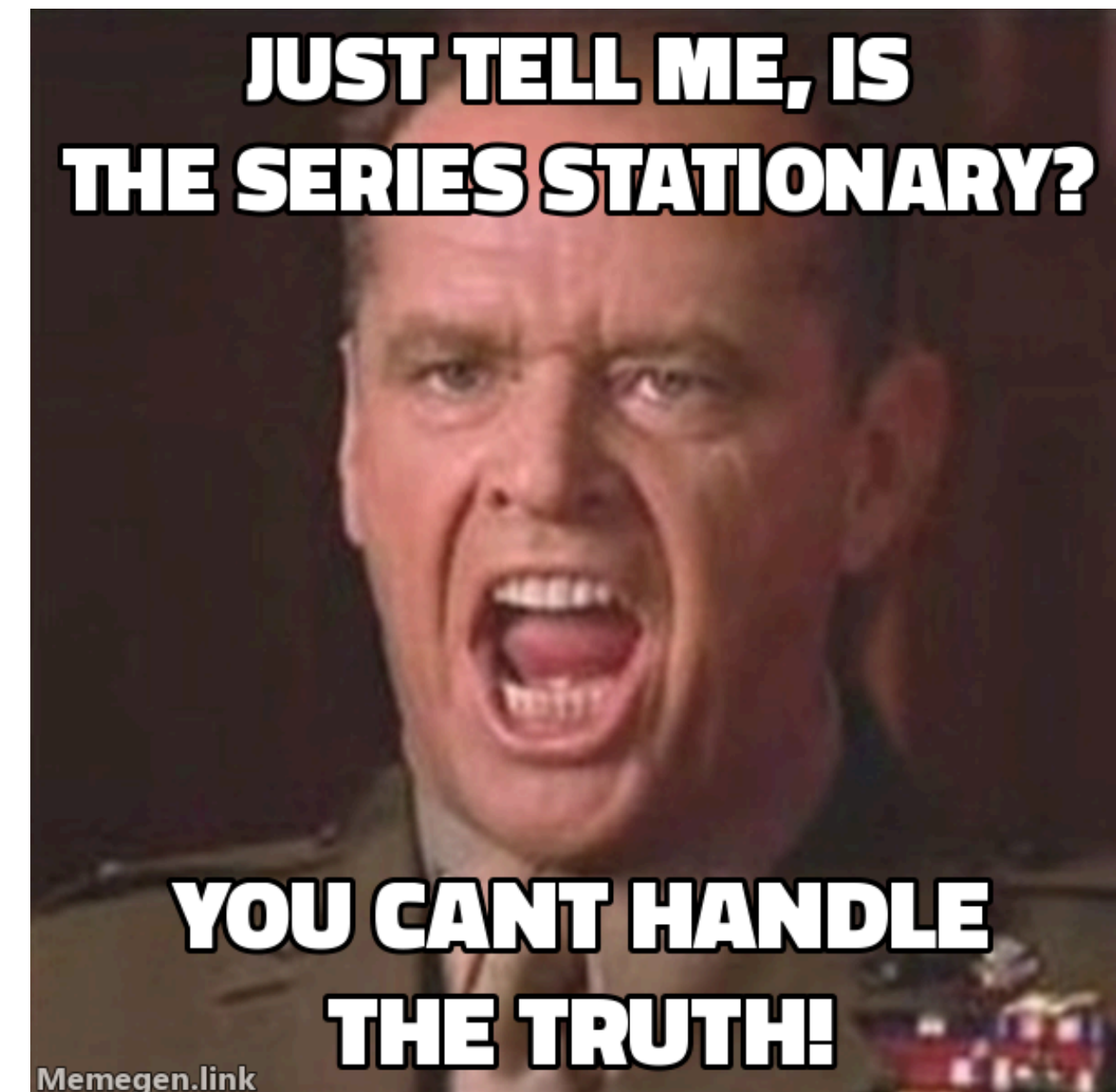
Prologue · The case reopens

Where we left off

Last module ended on a cliffhanger: two **independent random walks** handed us a beautiful R^2 and a fake-significant slope.

- $|\phi| < 1$: stationary, variance well-behaved.
- $\phi = 1$: random walk, variance grows without bound — spurious regression with anything.
- $\phi = 0.95$ vs. $\phi = 1.0$: nearly indistinguishable in small samples.

Today's question: given a series, *which side of the boundary are we on?*



Where we are on the mixing board

Module 2 turns on **no new sliders**. The master equation is the one from Module 1:

$$y_t = \alpha + \delta t + \sum_{j=1}^p \phi_j y_{t-j} + \sum_{l=1}^q \theta_l \epsilon_{t-l} + \epsilon_t.$$

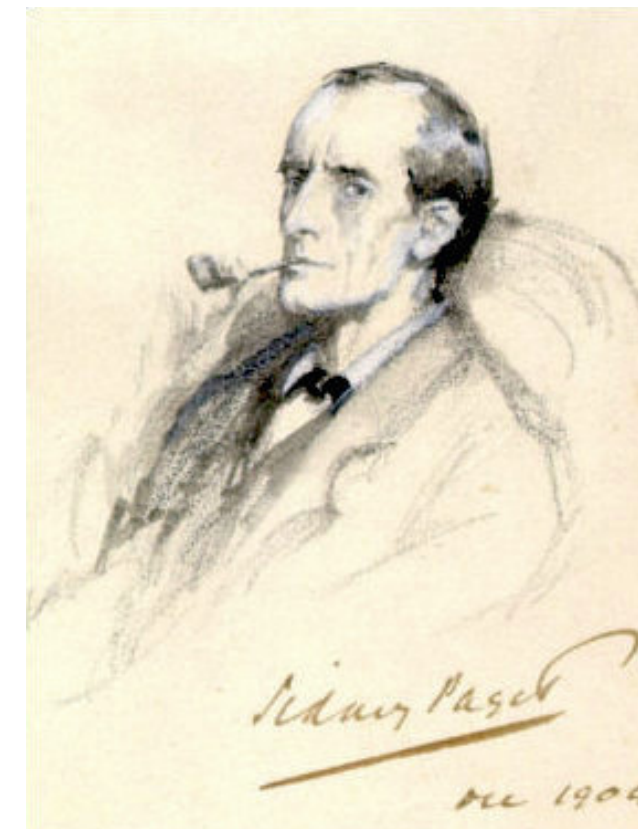
The DGP is still the AR(1) — α and ϕ_1 are the only active channels. What changes is the **question we ask of it**: is $\phi_1 = 1$?

One slider does make its first cameo: the deterministic trend δt . Deciding whether a trending series is trend-stationary or difference-stationary turns out to be part of the same testing problem — Act IV.

Stationarity is a verdict, not a measurement

Students want a one-test, one-number, one-answer protocol. Real data does not cooperate.

- No single piece of evidence is conclusive.
- We **build a case** from multiple sources — visual, ACF, formal tests, rules of thumb.
- When the evidence agrees, the answer is clear. When it conflicts, we lean on the **cost asymmetry**.



Sidney Paget, 1904. Deduction from evidence — under uncertainty.

The tool that does double duty: Δ

$$\Delta y_t = y_t - y_{t-1} = (1 - L) y_t, \quad \Delta^2 y_t = (1 - L)^2 y_t = y_t - 2y_{t-1} + y_{t-2}$$

- First difference = rate of change; second difference = acceleration. For economic series you rarely go past Δ^2 .
- Δ is both the **dependent variable** in the Dickey–Fuller regression *and* the **fix** for a non-stationary series.

Act I · The armchair

“You see, but you do not observe.”

ACT I – VISUAL INSPECTION



What the eye can and can't do

Three questions to ask any series you're handed:

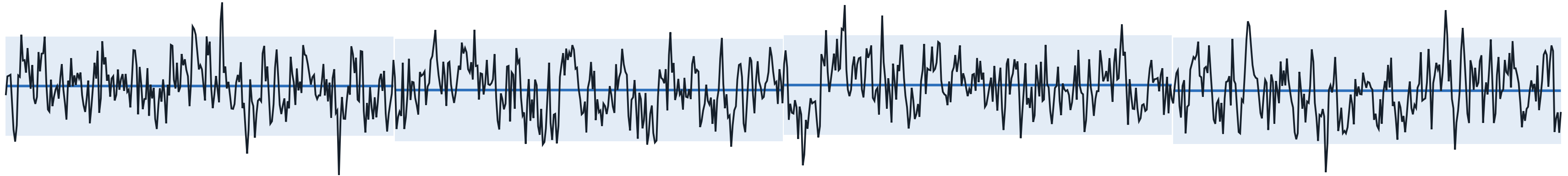
1. Does it **wander** without returning to a level? (**possible non-stationarity**)
2. Does the **variance** look like it changes over time? (**variance violation**)
3. Is there an obvious **trend**? (**then: trend-stationary or difference-stationary?**)

Bottom line. Visual inspection is the cheapest test we have — and the most easily misled. It is a starting point, not a verdict.

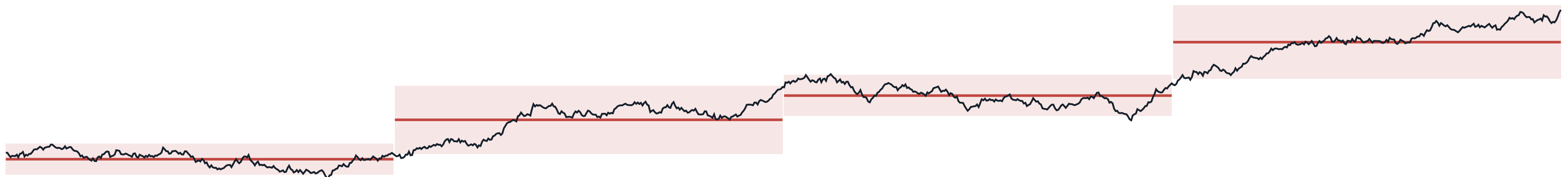
What stationarity looks like

Sub-period **means** (bars) and $\pm 2\sigma$ **bands** (shaded). Read them for the two things stationarity requires: a constant level and a constant spread.

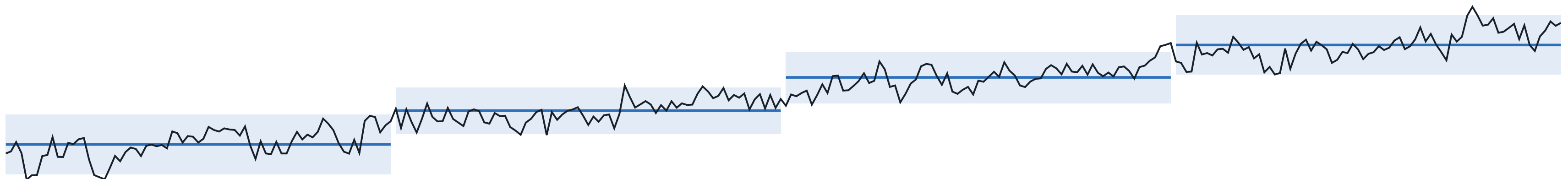
Stationary AR(1), $\phi = 0.5$ – sub-period means level, bands equal width



Random walk – means wander, bands widen



Trend-stationary: $y = 0.05 \cdot t + \text{AR}(1)$ – trends, but deviations are stationary



Top: means level off, bands equal width – *stationary*. Middle: means wander, bands widen – *random walk*. Bottom: the series trends, but the deviations from the line are well-behaved – *trend-stationary*, the classic trap.

Act II · The forensic lab

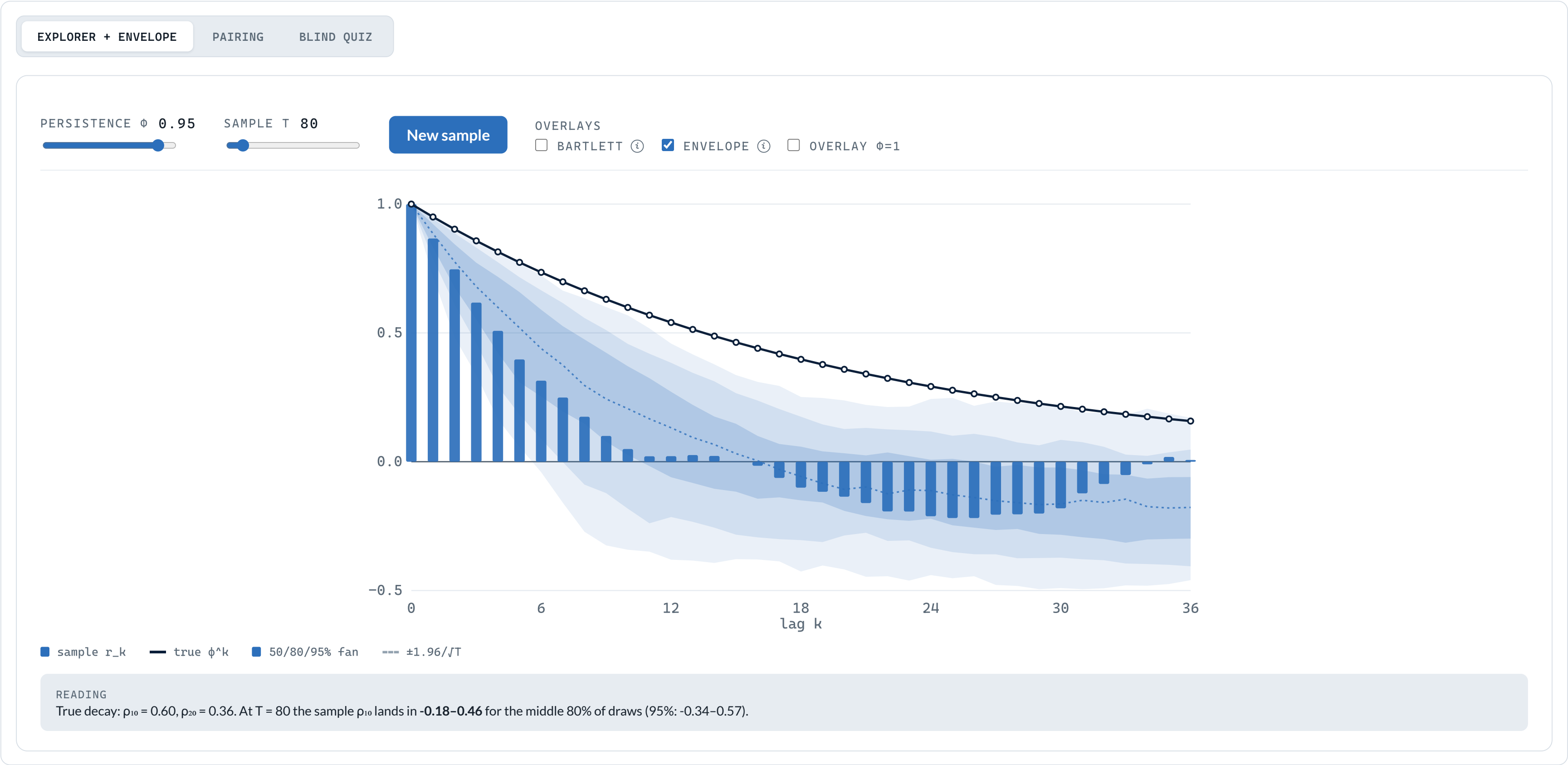
“The evidence doesn’t lie – but
you have to read it right.”

ACT II – THE ACF AND THE FORMAL TESTS



Exhibit A: the correlogram

You never see the $true \rho_k = \phi^k$ — only a noisy **sample** estimate. Crank T up and it converges; shrink T and push $\phi \rightarrow 1$ and it goes blind.



The instrument's blind spot

Flip the bench above to **Blind quiz**: you get *one* series and must call it — unit root or just persistent? At $\phi = 0.95$, $T = 80$ your hit rate hovers near a coin flip. **That blindness is why one witness is never enough** — and it's exactly where Act III begins.

What we're testing

Start from the AR(1): $y_t = \alpha + \phi y_{t-1} + \epsilon_t$.

- $H_0: \phi = 1$ (unit root, non-stationary) vs. $H_1: \phi < 1$ (stationary). One-sided.
- **First instinct that fails:** OLS t -stat on $\hat{\phi}$. Under H_0 , y_{t-1} is a random walk — standard asymptotics break, and the statistic is *not* t -distributed.

Building the DF regression

Subtract y_{t-1} from both sides and let $\gamma = \phi - 1$:

$$\Delta y_t = \alpha + \gamma y_{t-1} + \epsilon_t, \quad H_0: \gamma = 0 \quad (\phi = 1), \quad H_1: \gamma < 0.$$

- Under H_0 the LHS $\Delta y_t = \epsilon_t$ is stationary — the regression is at least well-behaved on the left.
- $\gamma = 0$ tests *exactly* the unit root. The **ADF** adds $\sum_{i=1}^p \beta_i \Delta y_{t-i}$ to soak up serial correlation.

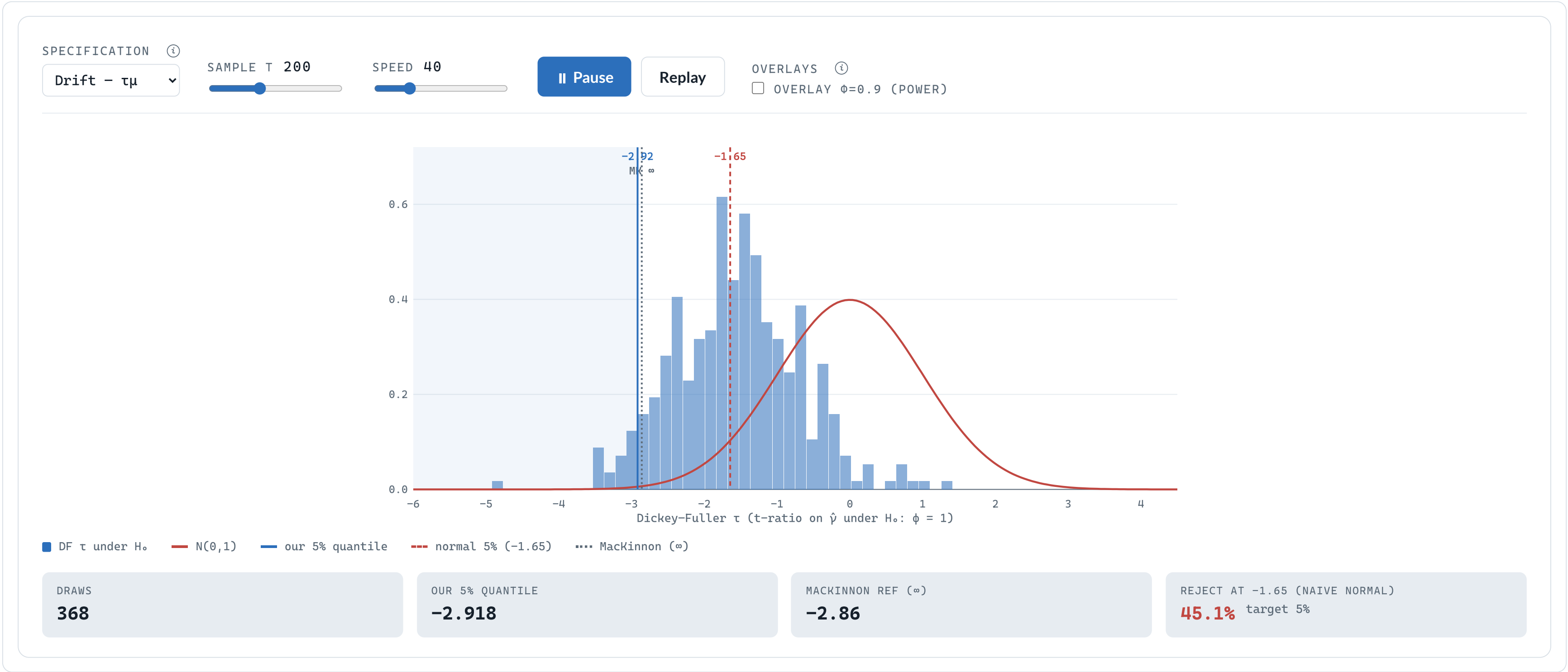
Three specifications — pick to match the plot

none: $\Delta y_t = \gamma y_{t-1} + \dots$ drift: $\Delta y_t = \alpha + \gamma y_{t-1} + \dots$ trend: $\Delta y_t = \alpha + \delta t + \gamma y_{t-1} + \dots$

Critical values differ across specifications — and adding deterministic terms *reduces power*. There is no free lunch. (See the widget on the next slide: watch the null distribution slide left as you add terms.)

Why a different null distribution? Watch it build.

The DF τ looks like a t -statistic, but its regressor is a random walk. We don't derive the limit — we **simulate it** and read the 5% cutoff straight off the pile.



Mackinnon critical values

$$CV_{0.05}(T) = -2.86154 - \frac{2.8903}{T} - \frac{4.234}{T^2} - \frac{40.04}{T^3} \quad (\text{drift}).$$

Mackinnon (1996) fit response-surface formulas to *exactly* the quantiles you just watched form — so you don't re-simulate every time. Critical values depend on T ; as $T \rightarrow \infty$, the drift 5% cutoff $\rightarrow -2.86$.

The function we keep using

MacKinnon's values aren't magic — they're the empirical quantiles of the pile you just watched form. One function, reused all semester:

```
1 mackinnon_cv <- function(T, level = 0.05, spec = "drift") {
2   if (spec == "drift" && level == 0.05)
3     return(-2.86154 - 2.8903/T - 4.234/T^2 - 40.04/T^3)
4   if (spec == "drift" && level == 0.01)
5     return(-3.43035 - 6.5393/T - 16.786/T^2 - 79.433/T^3)
6   if (spec == "drift" && level == 0.10)
7     return(-2.56677 - 1.5384/T - 2.809/T^2)
8   stop("Specification not implemented.")
9 }
```

At $T = 200$ this returns -2.876 ; our Monte-Carlo 5% quantile was ≈ -2.87 . **Same number, one from a formula, one from the pile.**

ADF from scratch

Strip the mystery: the ADF is one regression and one t -ratio. On a stationary AR(1) ($\phi = 0.7, T = 200$):

```
1 set.seed(8675309)
2 y <- arima.sim(n = 200, list(ar = 0.7)) # stationary
3 dy <- diff(y)
4 X <- embed(dy, 5) # [ $\Delta y_t, \Delta y_{t-1}, \dots, \Delta y_{t-4}$ ]
5 adf_reg <- lm(X[, 1] ~ y[5:(length(y)-1)] + X[, -1]) #  $\Delta y$  on  $y_{t-1}$  + 4 lags
6 adf_stat <- summary(adf_reg)$coefficients[2, "t value"] # t-ratio on  $y_{t-1}$ 
```

Quantity	Value
ADF τ -statistic	−4.92
5% MacKinnon CV ($T = 200$)	−2.88
Reject the unit root?	Yes (−4.92 < −2.88)

The packaged version

Every semester after: let the package do it. Same series, same answer.

```
1 library(urca)
2 adf_pkg <- ur.df(y, type = "drift", lags = 4)
3 summary(adf_pkg)           # reads off  $\tau_\mu$  and the critical values
```

Statistic	Value	1%	5%	10%
$\tau_\mu(\text{on } \widehat{\gamma})$	−4.92	−3.46	−2.88	−2.57

Identical to the by-hand τ , to the decimal — same regression, same t -ratio. Doing it once by hand demystifies the package — then we use the package.

Lag selection

How many lagged differences p ? A bias–variance trade:

- **Too few** → residual serial correlation leaks into the test and biases it.
- **Too many** → wasted degrees of freedom, lower power.
- `ur.df(..., selectlags = "BIC")` — our default for problem sets.
- **AIC** selects more lags than BIC — know that going in.
- **Schwert's rule:** start at $p_{\max} = \lfloor 12 (T/100)^{1/4} \rfloor$ and trim insignificant top lags.

KPSS: the reverse null

Test	H_0	H_1
ADF	unit root	stationary
KPSS	stationary	unit root

Opposite nulls → **confirmatory analysis**. Both have low power against close alternatives, so together they triangulate. When *both* fail to reject, the data genuinely cannot discriminate — an honest “I don’t know.”

KPSS in R

```
1 library(urca)
2 kpss_result <- ur.kpss(y, type = "mu")  # "mu" = level, "tau" = trend
3 summary(kpss_result)
```

Statistic	Value	10%	5%	1%
KPSS η_{μ}	0.080	0.347	0.463	0.739

0.080 < 0.463 → **fail to reject** stationarity, agreeing with the ADF. **Watch the polarity:** ADF rejects *for* stationarity at large **negative** values; KPSS rejects *against* it at large **positive** ones. This flips students up every semester.

The confirmatory grid — a reference card

The four outcomes, in one table (the live version is the Act III widget):

ADF (H_0 : unit root)	KPSS (H_0 : stationary)	Verdict
Reject	Fail to reject	Stationary
Fail to reject	Reject	Unit root
Reject	Reject	Conflict — break / borderline
Fail to reject	Fail to reject	Honest “I don’t know”

The last row is the one that matters: two tests of opposite nulls both failing means the data will not discriminate — usually a borderline ϕ or too small a sample.

Other tests — brief mentions

- **Phillips–Perron (PP)**. Same null as ADF, but a non-parametric correction for serial correlation instead of lagged differences. `urca::ur.pp()`.
- **Elliott–Rothenberg–Stock (ERS / DF-GLS)**. A more powerful ADF using GLS detrending — best exactly where standard ADF loses power (non-zero mean or trend). `urca::ur.ers()`. If you remember one alternative, remember this one.

None of them solves the fundamental power problem. Different tools, same job — which is why we still build a case.

Act III · Differential diagnosis

“Everybody lies – including your test statistic.”

ACT III – LOW POWER & THE BORDERLINE CASE



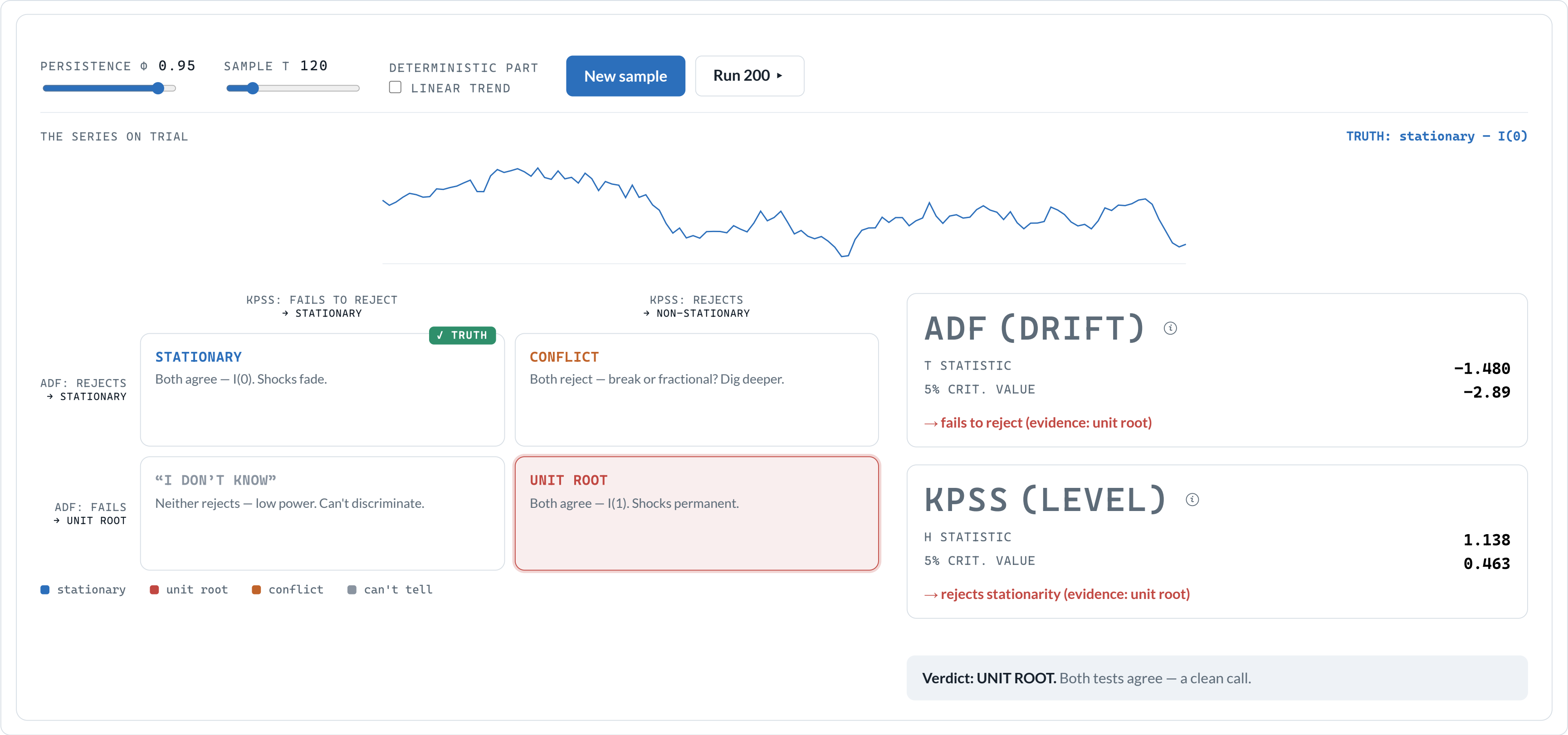
The single most important practical fact

Unit-root tests have **low power against highly persistent stationary alternatives**. At $\phi = 0.95$ in a short sample, the ADF often *fails to reject* even though the series is stationary.

- The borderline cases are exactly where testing is most useful **and** most fragile.
- The cure is not a better test — it's the **build-a-case** framework plus the cost asymmetry.

Two tests, opposite nulls — watch them disagree

ADF (H_0 : unit root) and KPSS (H_0 : stationary) together give **confirmatory analysis**. Crank ϕ toward 1, then **Run 200**: a *truly stationary* series lands on the right verdict only a fraction of the time. The scatter is the low power.



Act IV · The courtroom

“Beyond a reasonable doubt.”

ACT IV – THE VERDICT AND THE COST OF BEING WRONG



A rule of thumb

From Dr. Jeff Mills: compare the spread of the differences to the spread of the levels,

$$\frac{\sigma_{\Delta y}}{\sigma_y} \begin{cases} < 0.5 & \text{difference} \\ \geq 0.5 & \text{keep levels} \end{cases}$$

For a stationary AR(1), $\frac{\sigma_{\Delta y}}{\sigma_y} = \sqrt{2(1-\phi)}$: it's $\sqrt{2}$ at white noise and $\rightarrow 0$ as $\phi \rightarrow 1$. The < 0.5 cutoff flags $\phi \gtrsim 0.875$. Fast, informal — a sanity check, not a test.



Dr. Jeff Mills — pretty cool guy.

Why the ratio works

For a stationary AR(1),

$$\text{Var} \left(y_t \right) = \frac{\sigma_\epsilon^2}{1 - \phi^2}, \quad \text{Var} \left(\Delta y_t \right) = \frac{2\sigma_\epsilon^2}{1 + \phi}.$$



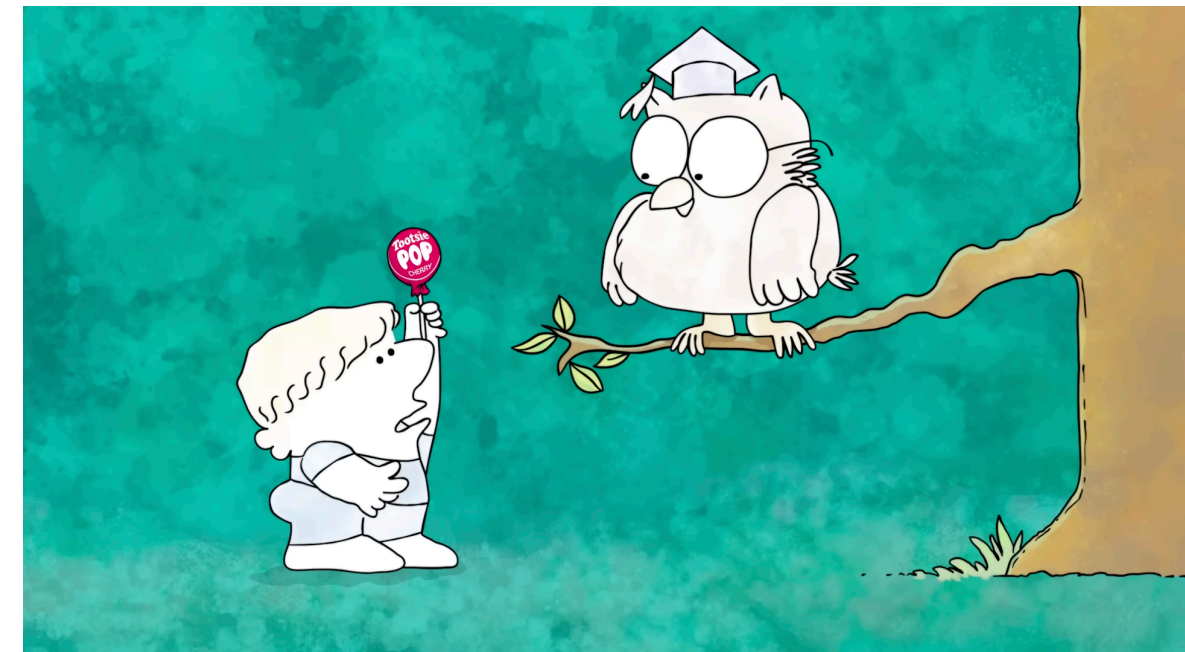
Their ratio collapses to

$$\frac{\sigma_{\Delta y}}{\sigma_y} = \sqrt{2(1 - \phi)}.$$

$\sqrt{2}$ at white noise, $\rightarrow 0$ as $\phi \rightarrow 1$. If differencing shrinks the spread a lot, the levels were accumulating shocks.

Order of integration & the over-differencing tax

$I(d)$ = number of differences to reach stationarity. $I(0)$ already stationary; $I(1)$ most common in economics; $I(2)$ rare. **Over-difference a trend-stationary series** and you introduce a non-invertible MA(1) ($\theta = -1$) — the tell-tale -0.5 spike at lag 1.



How many licks to the center? For economic series, almost never more than two.

The over-differencing tax, in algebra

Say the truth is trend-stationary: $y_t = \alpha + \delta t + u_t$ with u_t stationary. The *correct* move is to detrend. Difference instead:

$$\Delta y_t = (\alpha + \delta t + u_t) - (\alpha + \delta(t-1) + u_{t-1}) = \delta + \underbrace{(u_t - u_{t-1})}_{\Delta u_t}.$$

- If u_t is white noise, $\Delta u_t = \epsilon_t - \epsilon_{t-1}$ is an **MA(1)** with $\theta = -1$: $\theta(L) = 1 - L$, root on the unit circle $\rightarrow$ **non-invertible**.
- Poorly identified: ML pins $\hat{\theta}$ at -1 ; the ACF shows $\rho_1 = -0.5$. A unit root in the **MA** polynomial — just as bad as one in the AR, only hidden.

Detrend or difference? The wrong cure leaves a fingerprint

A trending series is either **trend-stationary** (detrend it) or **difference-stationary** (difference it). Apply *both* and read the correlograms: over-differencing injects the -0.5 lag-1 spike; under-differencing leaves a slow-decaying ACF.

TRUE PROCESS

Trend-stationary - $I(0)$ around a line

NOISE PERSISTENCE ϕ 0.00

SAMPLE T 150

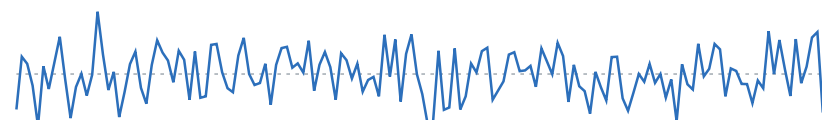
New sample

Truth: **trend-stationary** - stationary noise around a straight line. It is $I(0)$: detrend it.

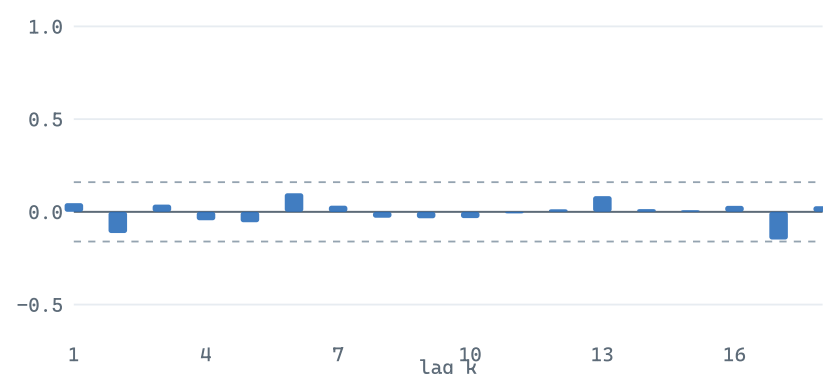
Detrend - remove $a + b \cdot t$

✓ CORRECT

RESIDUAL SERIES



CORRELOGRAM OF RESIDUALS

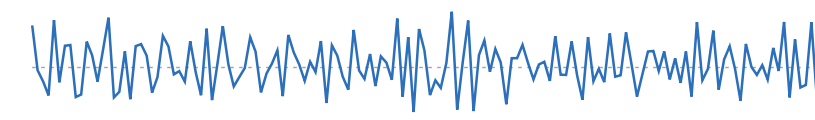


Residuals are the stationary part u_t - the ACF decays cleanly toward zero. **Right call.**

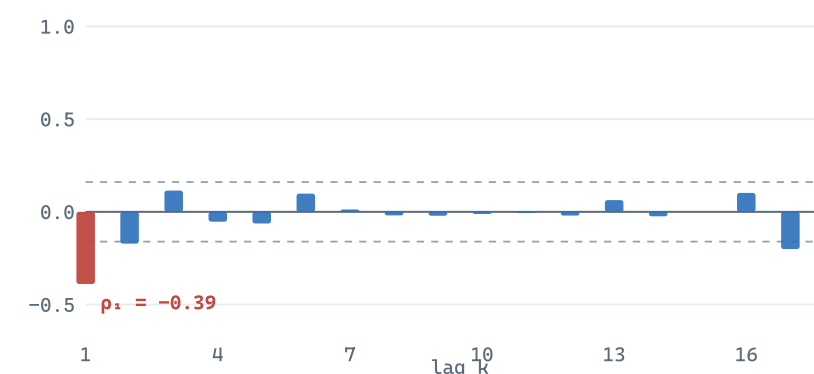
Difference - $\Delta y_t = y_t - y_{t-1}$

✗ OVER-DIFFERENCED

DIFFERENCED SERIES



CORRELOGRAM OF Δy_t



Differencing a trend-stationary series injects a non-invertible MA(1): the tell-tale $\rho_1 \approx -0.5$ spike. **Over-differenced.**

The four exhibits, weighed together

1. **Visual** — wander? variance change? trend?
2. **ACF** — fast decay (stationary) or slow/none (unit root)?
3. **Tests** — ADF (H_0 : unit root) + KPSS (H_0 : stationary), together.
4. **Rule of thumb** — $\sigma_{\Delta y} / \sigma_y$.

When all four agree, the verdict is clear. When they conflict — judgment, and the cost asymmetry.



The cost asymmetry

When in doubt, difference.

- **Over-differencing:** misspecified MA(1), but inference on coefficients of interest survives. *Detectable.*
- **Spurious regression:** a confidently wrong answer — big t , high R^2 , doesn't replicate. *The output looks good.*

The asymmetry is real. Spurious regression is the wrongful conviction you can't undo.



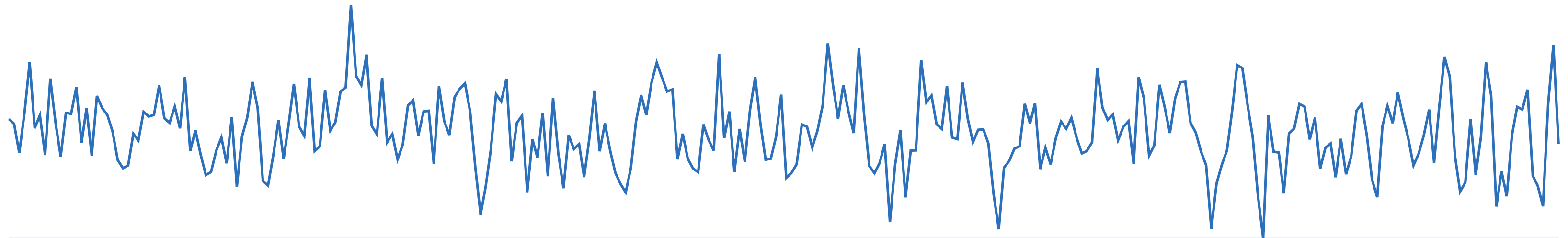
Practice: run the method yourself

We built one case together (a synthetic known-truth series, then FRED UNRATE — genuinely borderline). Now five fresh series, increasing difficulty, full evidence bundle each:

1. Easy stationary AR(1) — clear call.
2. Trend-stationary — choose the right ADF spec or misclassify.
3. Near unit root, short sample — honest “I don’t know” → difference.
4. Random walk with drift — a trend-looking series that isn’t trend-stationary.
5. FRED UNRATE — genuinely borderline; the lesson is the ambiguity.

Practice 1 · easy stationary AR(1)

Practice 1 – stationary AR(1), $\phi = 0.4$, $T = 300$

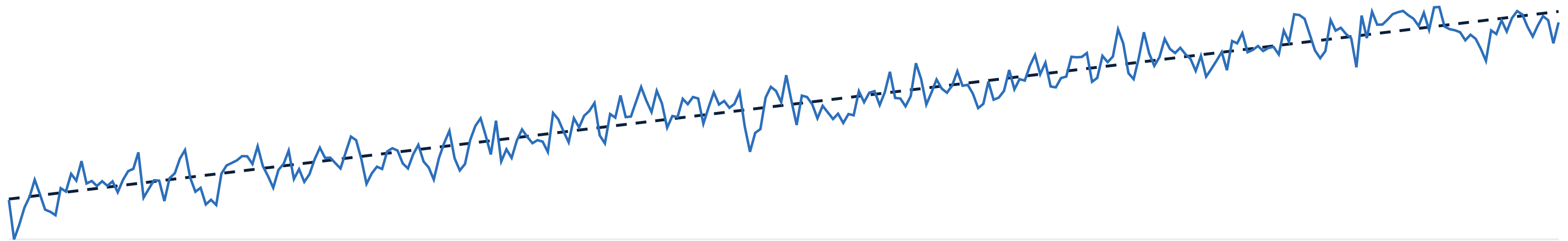


Evidence	Value	5% CV / cutoff	Reading
ADF τ_μ	-9.84	-2.87	Reject unit root
KPSS η_μ	0.265	0.463	Fail to reject stationarity
Ratio $\sigma_{\Delta y}/\sigma_y$	1.12	0.5	Well above cutoff

Verdict: Stationary. Easy call — all three agree.

Practice 2 · trend-stationary

Practice 2 – trend-stationary: $0.05 \cdot t + \text{AR}(1)$, $\phi = 0.5$

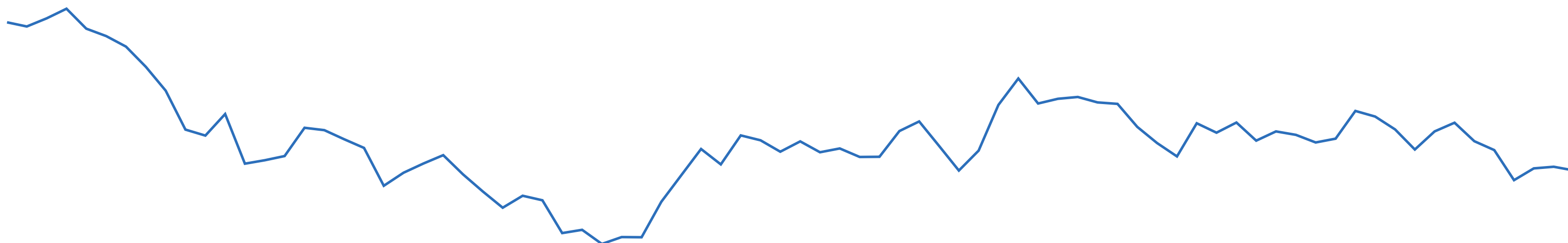


Evidence	Value	5% CV	Reading
ADF τ_{μ} (drift)	-2.30	-2.87	Fail – misreads the trend
ADF τ_{τ} (trend)	-9.37	-3.42	Rejects unit root
KPSS η_{τ} (trend)	0.077	0.146	Fail to reject trend-stationarity

Verdict: Trend-stationary. **Lesson:** pick the right ADF spec, or you misclassify.

Practice 3 · near unit root, short sample

Practice 3 – AR(1), $\phi = 0.95$, $T = 80$

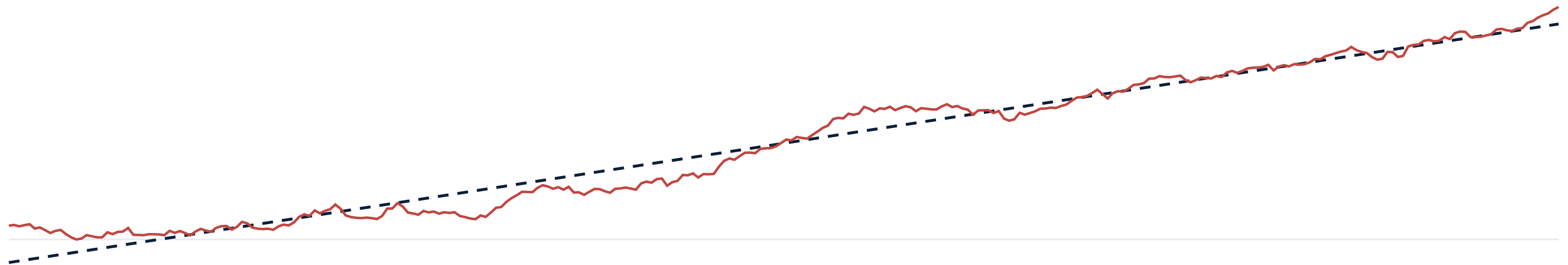


Evidence	Value	5% CV / cutoff	Reading
ADF τ_μ	-2.63	-2.90	Fail – low power
KPSS η_μ	0.317	0.463	Fail to reject
Ratio	0.37	0.5	Borderline

Verdict: honest “I don’t know.” Both fail – the $\phi = 0.95$ vs. 1.0 ambiguity. Apply the cost asymmetry: difference.

Practice 4 · random walk with drift

Practice 4 – random walk + drift

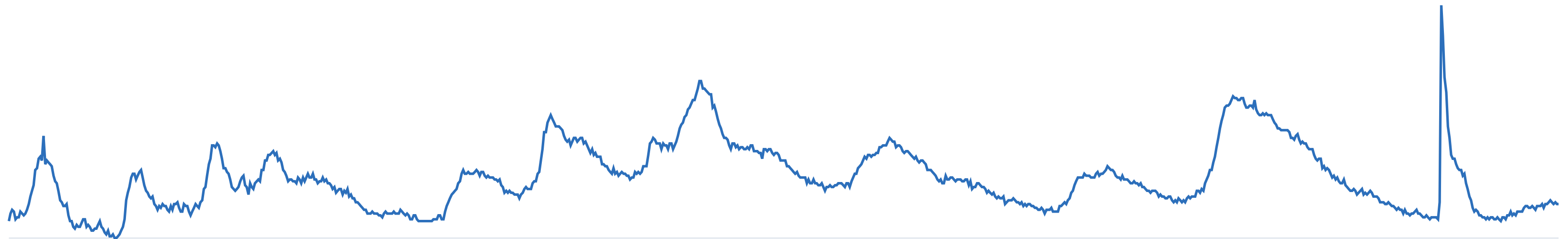


Evidence	Value	5% CV	Reading
ADF τ_{μ} (drift)	1.37	-2.87	Fail
ADF τ_{τ} (trend)	-2.98	-3.42	Fail – still can't reject
KPSS η_{τ} (trend)	0.515	0.146	Rejects trend-stationarity

Verdict: Difference-stationary. A trend-looking series need not be trend-stationary – the “trend” is a random walk with drift.

Practice 5 · FRED unemployment

Practice 5 – FRED UNRATE, 1948–present (%)



Evidence	Value	5% CV / cutoff	Reading
ADF τ_μ	−3.79	−2.86	Rejects unit root
KPSS η_μ	0.844	0.463	Rejects stationarity
Ratio	0.24	0.5	Favors differencing

Verdict: genuinely borderline — the tests **conflict**, and the series has decades-long excursions consistent with high persistence. The ambiguity is the lesson; apply the cost asymmetry and difference.

Key takeaways

1. Stationarity is a **verdict**, not a measurement — build a case from multiple sources.
2. The **DF regression** rearranges the AR(1) so the LHS is stationary under H_0 ; the stat is the t -stat on $\hat{\gamma}$.
3. Under H_0 that statistic is **not** normal — use **MacKinnon** values (they're just the quantiles you simulated).
4. **ADF + KPSS** (opposite nulls) give confirmatory analysis; both failing to reject = the data can't discriminate.
5. **Cost asymmetry**: spurious regression > over-differencing. When in doubt, difference — unless it's trend-stationary, then detrend.

Next time

- Beyond AR(1): general AR (p) and MA (q) .
- Reading the ACF **and** PACF together — the correlogram fingerprints of p and q .
- The Wold decomposition — why every well-behaved stationary series is AR + MA + noise.

Bring your laptop.